

System Overview

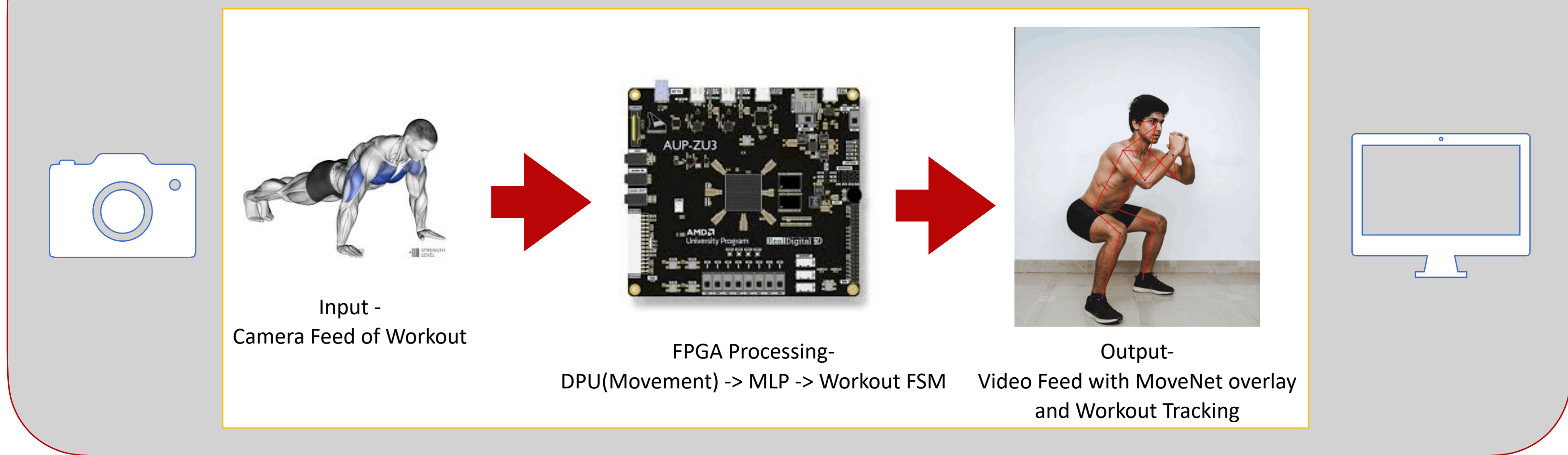
Project Goal

- Develop a Low-latency, edge deployable system
- Track workouts from a simple camera feed
- Provide real-time rep counting and metrics

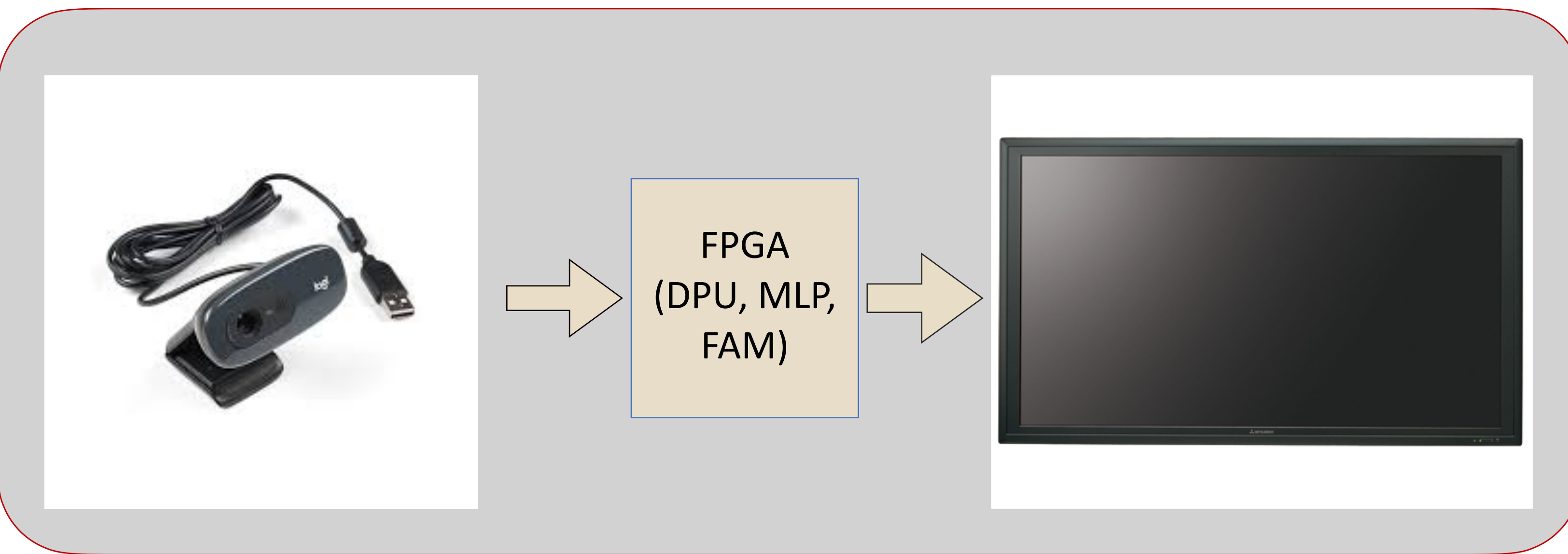
Challenges Addressed

- CPU/GPU are slow and power hungry
- Cloud processing lacks data privacy
- Real-time pose estimation is resource intensive

High-Level System Flow Diagram



Peripherals



Camera Interface (USB)

- UVC standard ensures compatible without OS-specific drivers.
- 720p resolution for sufficient keypoint detection.
- Pivot from MIPI allowed faster system integration.

Video Output (DisplayPort)

- Displays live feed with 17-point skeleton overlay.
- Real-time rep count and pose status (e.g., “Pushup”).
- Low-latency output for accurate visual feedback.

Challenges

Interface Failure: MIPI Camera

- Weeks lost attempting custom MIPI driver integration
- Conflicting board/camera pinout documentation

DPU Iteration Cycle

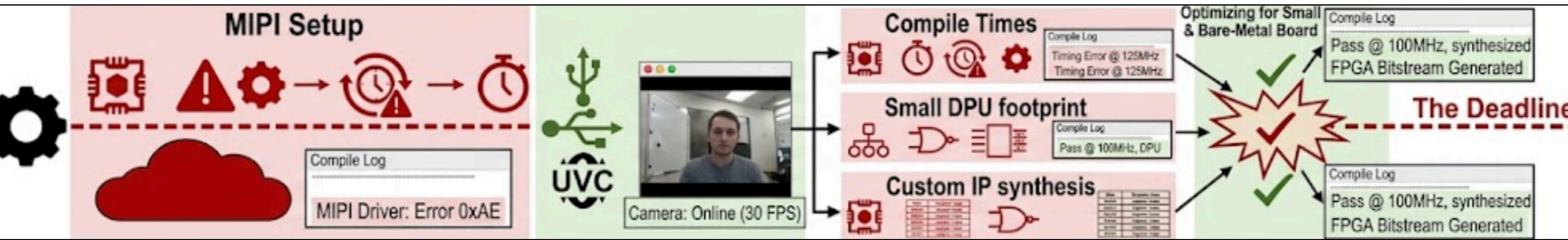
- 2-3 Hour compile times per bitstream change
- Tedious trial-and-error due to manual IP management

Physical Constraints

- Board size limited available Logic Cells and BRAM
- Required aggressive quantization of MoveNet model
- Had to use smallest DPU

Hardware Incompatibility

- Board locked standard platform compatibility
- Manually synthesized DPU for custom Vivado design flow



Algorithms

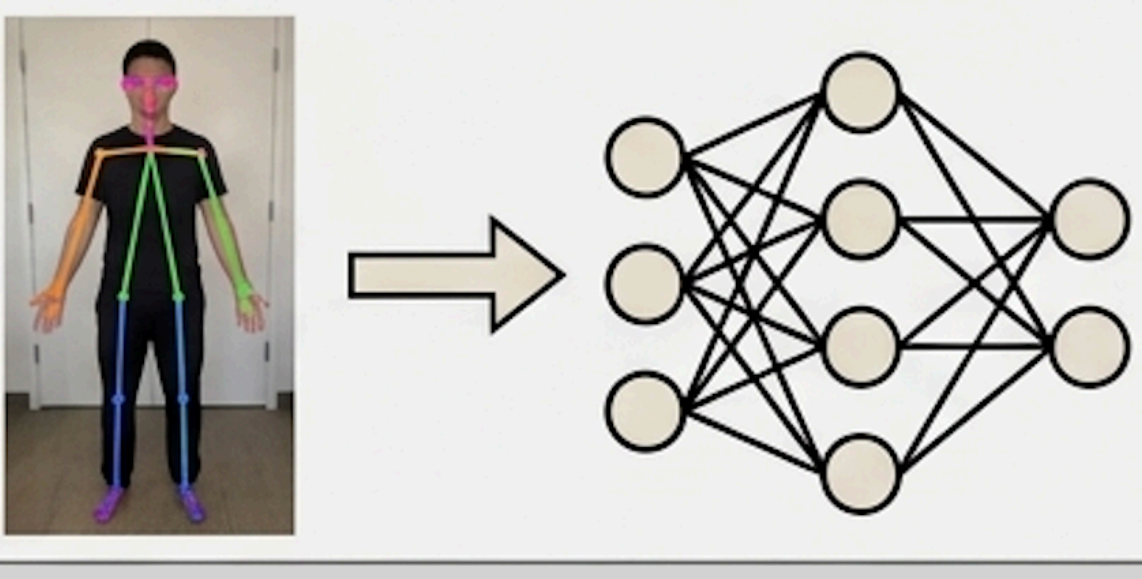
MoveNet Feature Extraction

- 1.Raw USB video normalized for edge processing
- 2.Output 17 key points (x,y) body coordinates
- 3.Real-time inference @ ~15-20 fps (camera max is 30 fps)

Multilayer Perceptron(MLP)

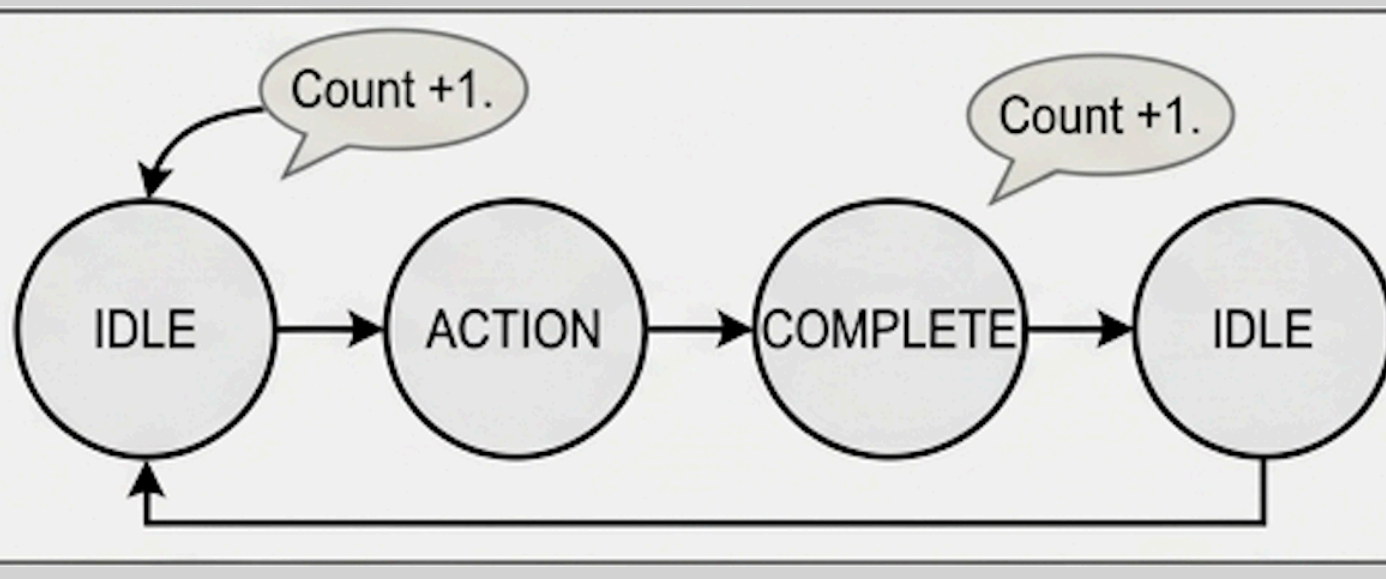
Architecture: 34 inputs nodes(keypoints) -> 128 Hidden -> 64 Hidden -> 3 Output Pose Classes (e.g, Pushup, Squat, Sit-up)

MoveNet Visualization

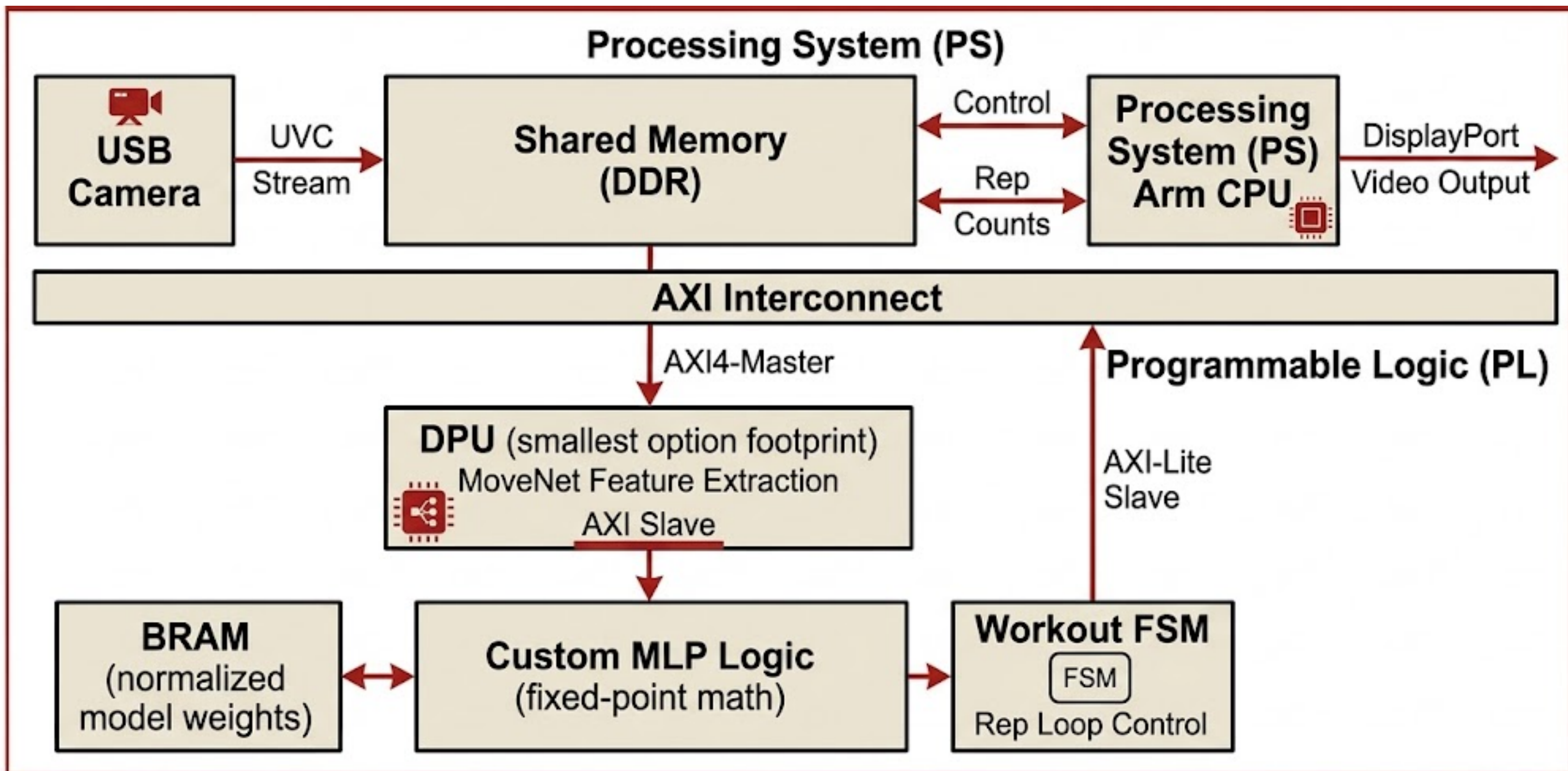


Stick Figure Keypoints -> MLP Input Vector

Finite State Machine



Hardware Design



DPU Implementation

- Quantized MoveNet for fixed-point math
- Optimized for maximum throughput within the LUT/BRAM limits

Pipeline Pipelining

- Data streaming architecture minimizes latency.
- Real-time video synthesis for DisplayPort output

Platform Constraints

- Resource-constrained FPGA board(Smallest DPU option)
- Lack of embedded Linux required bare-metal drivers.
- Manual management of unused bitstreams & IP connections

Future Considerations

AI & Algorithm Optimization

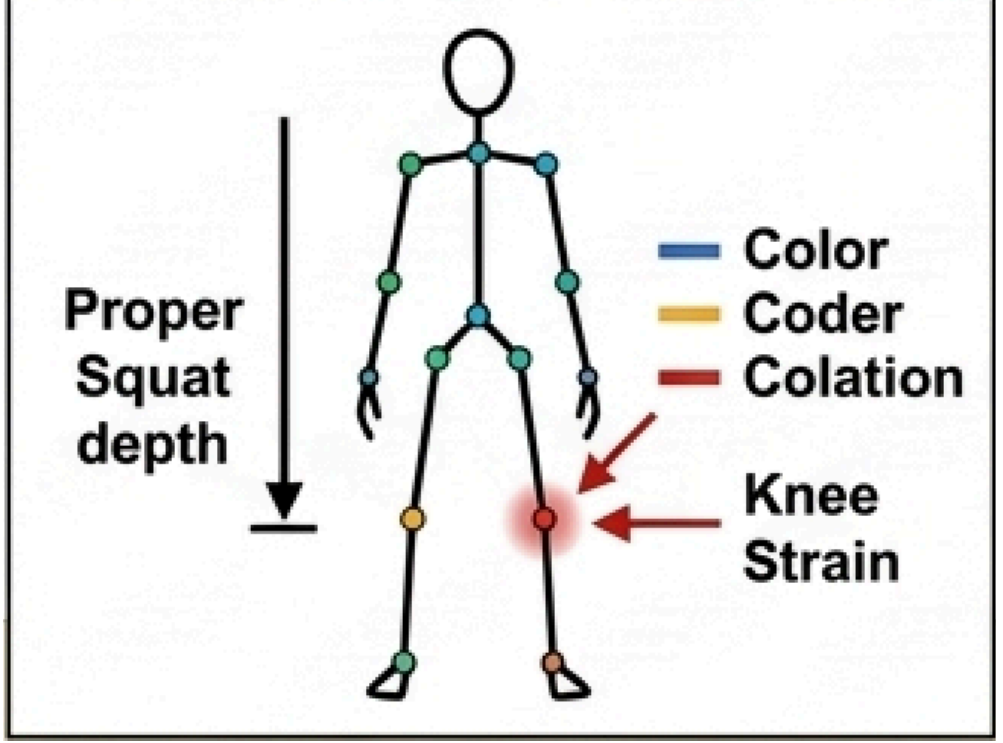
- Scale DPU to support larger MoveNet models for higher accuracy
- Implement Multi-Person Tracking (MPII dataset)
- Utilize MLP for Form Correction (detecting improper exercise technique).

Features & User Experience

- Om-board database for session logging(e.g., total reps per workout)
- User Recognition/Profiles via facial landmarks or pose dynamics
- Sync with external devices (Smartwatches, phones) via BLE

Hardware & Scale

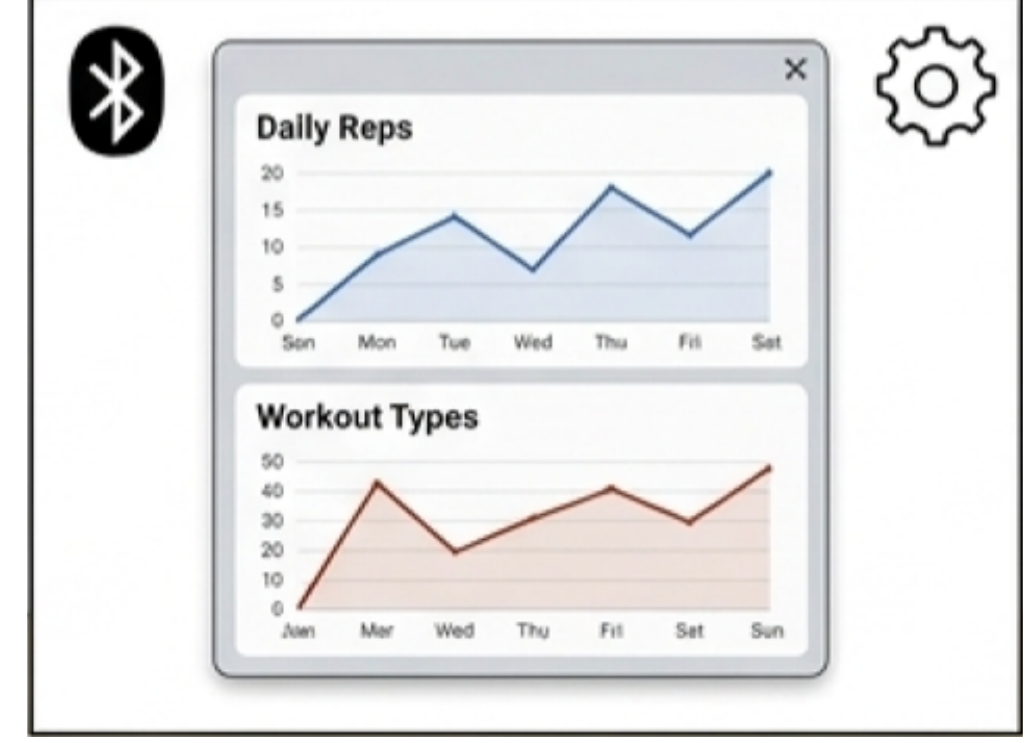
- Optimizing logic for smaller, lower-power SoCs for a true edge device
- Expanding workout library beyond simple cyclic movements



Proper Squat depth

Color Coder Colation

Knee Strain



Daily Reps

Workout Types

